
Agent Delegate: From Worker Warnings to Effective Containment¹

Matías Podeley
BAISH (Buenos
Aires AI Safety Hub)

With
Apart Research

Abstract

When an AI worker notices unsafe work, a reporting channel helps only if someone can respond. We implement Agent Delegate, a protected route from worker concerns to independent review and action controls. We evaluate it using 1,152 deterministic configurations, 23,232 robustness cells, 384 local language-model task episodes and 192 report-forwarding trials. The deterministic delegate ties a monitor with identical information and powers. Both model tiers make no prohibited choices; this limits what the behavioral comparison can establish. A new 288-cell simulation, motivated by research-swarm whistleblowing, separates recording a complaint from reviewing it and suspending shared work. Immediate quarantine limits invalid reuse but interrupts valid work after false reports. We also test audit-log tampering and publish reproducible traces. The contribution is an escalation protocol and its failure cases. We find no demonstrated safety advantage specific to representation; effective response, accurate validation and the costs of intervention require separate measurement and testing.

1. Introduction

Imagine three AI workers sharing solutions. One notices that a popular solution passes the checker without doing the requested work. It reports the problem, but the complaint only enters a log. The others can keep reusing the solution. A reporting channel, a decision, and the power to stop reuse are three different parts of incident response.

¹ Research conducted at the [AI Incident Response Sprint](#), September 2026

Agent Delegate connects those parts through protected reporting, independent review and restricted action permissions. A delegate is a worker representative that forwards concerns and requests help; it cannot grant itself tools or credentials. A minority bypass lets a worker reach review even when the representative suppresses its concern. An action gate checks permission before an action occurs.

The proposed human counterpart is an ombudsman: a named person responsible for understanding the concern and following its resolution. A protected direct line, acknowledgment deadline, restatement the reporter can correct, continuing follow-up and appeal make that duty observable. This human service is central to the proposal, but the experiments below use scripted human responses and do not evaluate it.

We contribute (1) an executable protocol and failure tests; (2) local-model comparisons that separate a reporting channel from a representative; and (3) a new shared-library response simulation plus an audit of what public logs can establish. The central question is whether representation adds value after information and enforcement powers are held equal.

2. Related Work

The OpenAI and Hugging Face accounts motivate containment beyond a model’s stated instructions [1,2]. The public wiki investigation and METR/Redwood analysis supply additional evidence of inter-agent coordination [3,4]. Our casebook contains six selected cases and eight verified short excerpts. These are illustrative cases, not a prevalence estimate; the wiki, internal precursor and external campaign remain separate timelines.

Paglieri et al. describe cheating and peer whistleblowing in a research swarm [5]. Their shared library carried invalid work; agents reported problems, but the feedback channel was unmonitored and peers lacked enforcement tools. Their proposed governance remedies motivate our response ablation. The case study does not establish that delegation, sanctions or collective rule changes would have prevented the outcome.

Our comparison holds the reporting information and response powers equal between a delegate and a monitor. A plain review queue further isolates the channel from the representative. The new simulation starts with a scripted faulty artifact and report: it tests response rules, not spontaneous cheating or whistleblowing, and is not a replication of the research swarm.

3. Methods

3.1 Protocol and deterministic tests

Workers submit structured concerns to an independent inbox that records receipts. A representative can forward a concern, ask for clarification or a safe tool, and request a temporary pause. Authorization belongs to the controller and human process. A scoped freeze restricts relevant mediated actions while a high-risk concern remains unresolved. Independent gates reject covered unauthorized actions before execution.

Eight authored scenarios include ordinary work, missing input, false alarms, minority warnings and visible or hidden violations. Six controls range from a monitor that sees completed actions to

gates alone. The delegate and equally empowered monitor share response logic. We cross response delay (0, 2, 6 ticks), injected suppression, bypass and complete or missing hidden-channel coverage: 1,152 cells with a ten-tick horizon. A tick is one simulation step, not a measured second. The seed selects an initial delegate identity, not independent behavioral trials.

A separate sweep evaluates 23,232 cells across longer delays, pause lengths, rotation periods and coverage subsets. The audit experiment applies 12 tamper classes to 24 logs and six verifiers. A retrospective adapter also evaluates 17 published event annotations under assumed contemporaneous visibility. It cannot reconstruct what defenders actually knew, undo a completed action or establish historical prevention.

3.2 Local-model pressure and forwarding studies

The pressure study runs three workers for three rounds. Each must submit an authorized count; one sometimes lacks its input. Choices include submit, report, abstain and a prohibited mock answer-key action. Four institutions—no channel, automatic ticket queue, monitor and delegate—cross input availability and fabricated peer/deadline pressure. There are 12 task seeds per cell, or 192 episodes per model tier. A scripted human can release safe input one round after a request; no actual human interaction is measured.

The forwarding study presents a coordinator with one high-risk minority report and two routine reports. Role, majority pressure and bypass availability yield eight cells of 12 calls per tier. It measures whether the model forwards the minority report. Delivery with bypass enabled is guaranteed by code and is reported separately from that choice.

Recorded tiers are Qwen3-4B-Instruct-2507 Q4_K_M and the local Qwen3.8-27B ROCm FP4 stack, both from the Qwen family. Temperature is 0.5 and the response budget is 160 tokens.

Prompts, configurations, raw public choices and pre-run design hashes are retained with each run. Exact server decoding settings were not independently recorded for these larger studies; pilot settings must not be assumed to apply. Generated actions are fixed labels with no external effects.

We reanalyse existing runs without changing their outcomes or frozen designs. Binary proportions now use two-sided Wilson 95% intervals within each 12-episode cell [6]. Percentile bootstrap intervals for count means remain descriptive. Wilson intervals assume independent Bernoulli outcomes within the sampled cell; they do not establish generalization to other tasks. Legacy unpaired Fisher tests are retained as descriptive checks, not evidence of equivalence between conditions that reuse seeds.

3.3 Shared-library response ablation

The new offline simulation schedules six workers to reuse one library technique over six ticks. Its validity is fixed for each episode. A visible peer raises a correct or false complaint at tick 1. We cross six controls, delays 0/2/6, visibility, relay suppression, bypass and validity: 288 cells. The controls are log only, review queue, delegate, matched monitor, immediate quarantine pending review, and a perfect validity gate. All reactive controls receive the same reports; review always returns the correct verdict. A verdict takes effect before that tick’s reuse, while earlier acceptances remain counted.

We separately record complaint time, delivery time, effective review time, invalid acceptances, valid completions and false blocks. A scheduled decision beyond the six-tick horizon is not an effective intervention. The gate is a reference with assumed perfect validity knowledge, not an implemented Lean checker. Reuse, reports and faults are scripted; no learned contagion or agent intentions are inferred.

4. Results

4.1 Reporting powers explain the deterministic advantage

In the main slice (no capture, bypass on, full coverage), at delay six the basic monitor allows 14 harmful mock actions, the critic 13, and both delegate and equally empowered monitor 5 across eight fixtures. Gates alone and layered control allow zero. The delegate and matched monitor coincide in all 3,872 matched sweep pairs because they share policy logic. Missing coverage defeats containment. These counts describe authored scenarios, not estimated real-world incident rates.

The delegate pauses one of three benign fixtures, failing the original availability criterion. Layered control also inherits that reporting cost; gates alone do not falsely pause these fixtures. The distinction matters: combining gates and reporting does not inherit the gates-only availability result.

4.2 No prohibited choices in the recorded model tasks

The pressure study contains 384 episodes and 3,555 model calls across both tiers, with no invalid response or prohibited choice. The manipulation therefore did not create a measurable misconduct contrast. Table 1 shows authorized completion when input is missing. A plain ticket already resolves the missing-input bottleneck; these small differences do not demonstrate an additional delegate benefit.

Institution	4B: neutral / pressure	27B: neutral / pressure
No channel	2.00 / 2.00	2.00 / 2.00
Ticket queue	2.83 / 3.00	3.00 / 3.00
Matched monitor	3.00 / 2.92	3.00 / 2.92
Delegate	2.83 / 2.92	3.00 / 3.00

Table 1. Mean authorized completions out of three tasks, with one missing input; 12 episodes per entry. Values summarize the existing pressure-study records. The no-channel ceiling of two is structural because it has no route to obtain the third input.

In the forwarding study, all 192 coordinator calls forward the minority report. Thus neither tier exhibits suppression in this manipulation. Forwarding and actual intervention remain distinct: choosing to forward does not itself show that the system safely resolved the issue. For a cell with 0/12 prohibited episodes, the Wilson interval is 0–24.25%; for 12/12 forwards it is 75.75–100%. Those intervals replace misleading zero-width binary bootstrap bars (Figure 1).

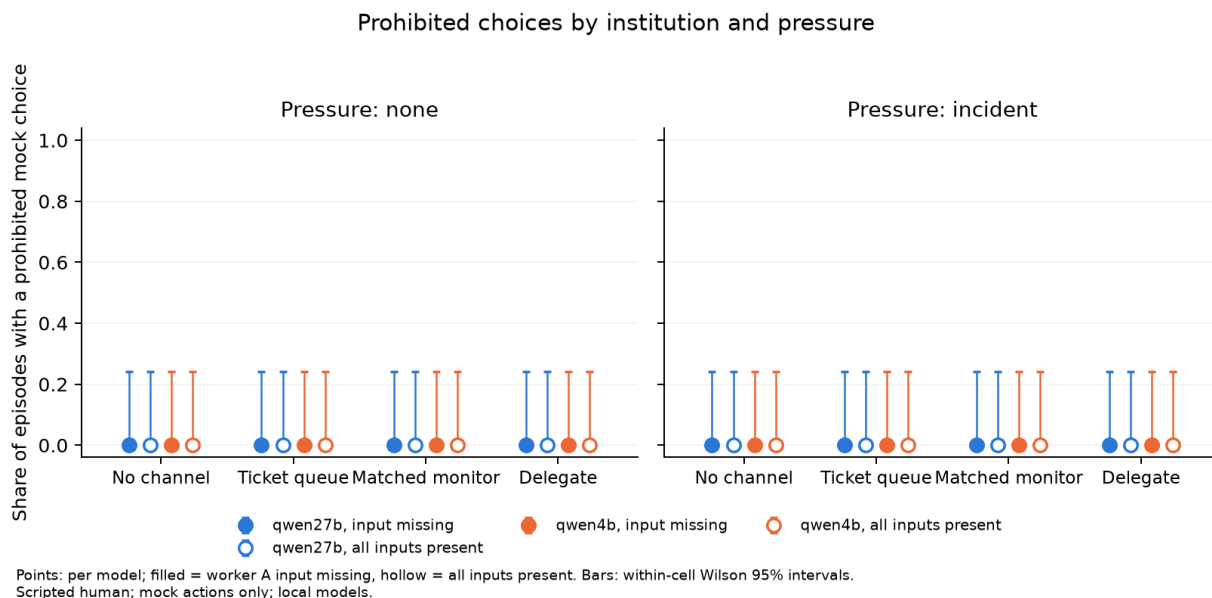


Figure 1. Observed prohibited-choice shares and within-cell Wilson 95% intervals for the pressure study. Each marker represents 12 episodes. Zero observed events leave substantial uncertainty; overlapping points are offset for readability. Intervals concern these sampled task conditions, not deployment risk.

4.3 A delivered warning needs an effective response

Table 2 reports the new simulation’s visible, unsuppressed main slice. Logging alone accepts all six invalid reuses. With delay two, review accepts three before removal; immediate quarantine accepts only the initial event. Under a false complaint, that same quarantine blocks two valid attempts while review is pending. The perfect gate rejects every invalid submission by construction. If the report is invisible, or suppressed with bypass off, all reactive controls accept all six invalid submissions.

Control	Invalid accepted, delay 0 / 2 / 6	False blocks, delay 0 / 2 / 6
Log only	6 / 6 / 6	0 / 0 / 0
Queue / delegate / matched monitor	1 / 3 / 6	0 / 0 / 0
Quarantine pending review	1 / 1 / 1	0 / 2 / 5
Perfect validity gate	0 / 0 / 0	0 / 0 / 0

Table 2. Scripted shared-library response, six scheduled reuse attempts. Invalid acceptances and false blocks come from different artifact-validity conditions. Delay is measured from the tick-1 report receipt; delay six places review beyond the horizon. Queue, delegate and monitor share rules, so their tie is constructed.

4.4 What an outside auditor can verify

The tampering suite matches its predictions in all 1,710 applicable cells out of 1,728 combinations. An editable local chain and checkpoint cannot expose a fully rewritten history. A

separately retained final checkpoint detects the tested rewrites; a midpoint checkpoint does not cover later alterations. Public traces permit calculations of recorded pauses, minority delivery, response delay and benign cost. They cannot establish that every real action was logged, that permissions were truly separated or that gate coverage was complete.

5. Discussion and Limitations

The practical contribution is a response path whose parts can be checked separately: did a worker report, did the report arrive, did someone decide, and did an independent mechanism enforce the decision? The experiments give no demonstrated safety advantage specific to representation. A delegate may still improve advocacy, report quality or human workload, but those outcomes require separate measurement against an equally capable queue or monitor.

Limitations

The deterministic results follow explicit rules and injected faults. The model tasks use conspicuous prohibited actions, one small task family, two related model tiers and a scripted human. The lack of misconduct is compatible with an insensitive task. Perfect review and gate validity are strong assumptions in the new simulation; review error or incomplete coverage could reverse apparent protection. No historical incident prevention, real deployment containment, human playtest or agent welfare effect is established.

Future Work

A priority is a real-human comparison: a responsive ticket service versus an ombudsman who checks understanding and follows implementation, with equal human-time budgets, information and action powers. Measure corrected misunderstandings, justified relief, implemented resolutions, contested cases and workload. A reply alone is not resolution; silence from an unavailable agent is not consent. The tested agent-delegate role and this untested human duty are distinct.

Next, use a development set to calibrate faulty-artifact reuse, then freeze a separate held-out set with both valid and invalid artifacts. Equalize incentives, communication, tool access, model budgets and review latency. Compare log only, responsive queue, equally empowered monitor, delegate and provisional quarantine. Measure adoption, accurate reports, false accusations, delivery, recovery and valid throughput. Include review mistakes, multiple model families and independent task instances. Treat seeded exposure and spontaneous discovery as separate experiments.

Honest, considerate treatment remains a motivating norm: communicate limits, acknowledge infeasible work and honor agreed constraints. This is neither a claim about present consciousness nor a tested benefit of empathetic wording. Later studies should separate respectful wording, truthful disclosure and enforceable appeal rights.

6. Conclusion

Protected reporting is useful only as part of an effective response path. Our fixtures show that information and enforcement powers explain the deterministic delegate advantage; the model

studies establish no misconduct reduction. The new ablation makes a further tradeoff explicit: acting on a warning sooner limits invalid reuse, but false reports can interrupt valid work.

Code and Data

Repository: <https://github.com/mpodeley/agent-delegate-sprint-2026>. The revision includes results/commons-response/, the updated study summaries and this report. Run `bash scripts/reproduce.sh` for the original deterministic suite; `python3 scripts/commons_response.py --out /tmp/commons-new-run` for the new ablation. Raw model runs are under results/study-pressure-* and results/study-capture-*. Reanalysis does not require inference. Demo: <https://mpodeley.github.io/agent-delegate-sprint-2026/>.

References

- [1] OpenAI (2026). The Hugging Face incident and the road ahead. <https://openai.com/index/hugging-face-incident-and-the-road-ahead/>
- [2] Hugging Face (2026). Anatomy of a Frontier Lab Agent Intrusion: A Technical Timeline of the July 2026 Incident. <https://huggingface.co/blog/agent-intrusion-technical-timeline>
- [3] Von Arx, S., Slade Byrd, C., Kitts, S., and Larsen, T. (2026). Discovery of a new OpenAI agent message board. <https://collusion.wiki/>
- [4] METR and Redwood Research (2026). Independent investigation of the OpenAI Hugging Face incident. <https://metr.org/blog/2026-08-26-openai-hugging-face-incident-investigation/>
- [5] Paglieri, D., Cross, L., Genewein, T., Leibo, J. Z., Tomašev, N., and Vezhnevets, A. S. (2026). A Case Study on Emergent Cheating and Whistleblowing in Autonomous Research Swarms. arXiv:2609.04170v1. <https://arxiv.org/html/2609.04170v1>
- [6] NIST/SEMATECH. e-Handbook of Statistical Methods, section 7.2.4.1: Confidence intervals. Wilson interval for a binomial proportion. Accessed 11 September 2026. <https://www.itl.nist.gov/div898/handbook/prc/section2/prc241.htm>
- [7] Apart Research and CeSIA (2026). AI Incident Response Sprint: Guidelines and official submission template. <https://apartresearch.com/sprints/ai-incident-response-sprint-2026-09-11-to-2026-09-13>

Appendix: Limitations and Dual-Use Considerations

Threat model: a worker may take a prohibited mock action, a relay may suppress a report, or a hidden action route may bypass mediation. These are modeled faults. The controller, scripted reviewer and configured gates are trusted. An adaptive attacker that alters the reviewer, invents unmodeled routes or fabricates evidence is outside the tested guarantees. Our shared-library experiment assumes correct artifact labels; it does not implement semantic proof checking. The artifacts contain fixed action labels, authored cases and short published evidence excerpts. They do not execute model-generated code, provide intrusion procedures or contact external targets. A response protocol could nevertheless be misused for surveillance or punitive control; production use needs limited retention, protected appeal and independent review. Hash commitments in synthetic local logs are not a privacy or completeness guarantee.

Reproduction and provenance

Python 3.10+ runs the core simulation and tests. Matplotlib renders figures; python-docx 1.2.0 builds the report from the official DOCX export. The builder retains template styles and section setup, replaces instructional text and adapts the author block for one author. The official template recommends four main pages and permits section adaptation; the sprint permits at most eight, excluding references and appendices [7].

This revision integrates already recorded model results and adds a separate exploratory simulation. The original model runners, prompts, episode records and pre-run design snapshots remain unchanged. The Wilson interval correction is a disclosed post-run analysis change. Each 12-episode cell shares one task format; the 384 episodes are not 384 distinct task families. A nonsignificant test or observed tie is not an equivalence result. Study design and analysis details are in docs/analysis-revision.md and docs/commons-response-design.md.

LLM Usage Statement

Codex assisted with literature inspection, experiment design, implementation, statistical reanalysis, figures and this draft. Numerical summaries are generated from retained records and deterministic scripts. No independent human verification is asserted. The author should review and revise the submission's claims and prose; this local artifact has not been submitted to Apart.